

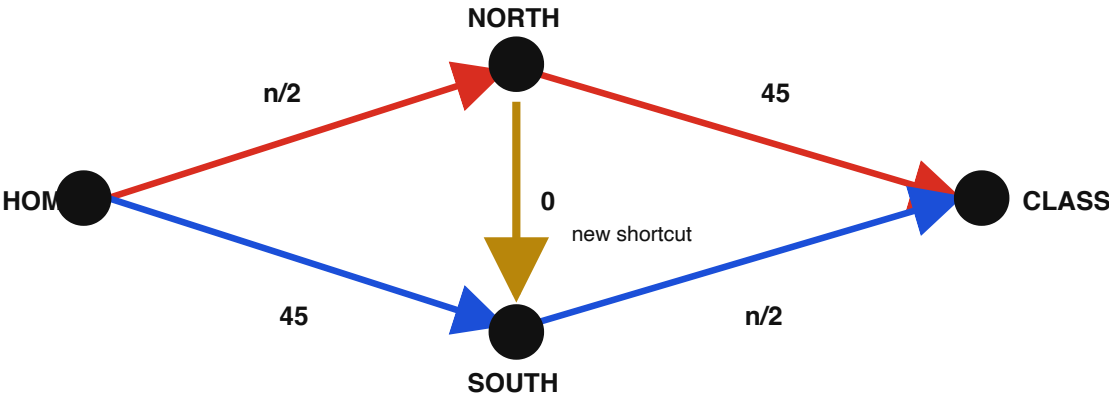
How systems behave

Jumpstart 2026 - Day 1 field guide - keep this for tomorrow

The map

68 commuters

Every number is travel time in minutes. On a road marked $n/2$, n is the number of people using that road.



Round 1

red or blue

ROUTE	PEOPLE	TRAVEL TIME
Upper		$45 + R/2 =$
Lower		$45 + B/2 =$

Would anyone improve their own trip by switching routes?

Round 2

red, blue, or both

ROUTE	PEOPLE	TRAVEL TIME
Upper		$45 + (R + S)/2 =$
Lower		$45 + (B + S)/2 =$
Shortcut		$34 + S/2 =$

Would anyone improve their own trip by switching routes?

A different question

What is best for all 68 commuters?

Use the calculus you already know

6 min - at your table

Let y people use the shortcut. Split everyone else evenly between the two ordinary routes.

ORDINARY ROUTE

$$62 + y/4 \text{ minutes}$$

SHORTCUT ROUTE

$$34 + y/2 \text{ minutes}$$

1. Write the total travel time, $T(y)$, for all 68 commuters.

2. Expand it. Then find $T'(y)$.

3. What value of y minimizes total travel time?

4. How many people take each route?

What bothers you?

write before we talk

Is closing a road a loss of freedom if everyone gets home sooner?

What should an engineer optimize: the average, the worst trip, or fairness?

When reasonable choices produce a bad system, who is responsible?

The worry wall

anonymous - before you leave

First colour: one worry about first year at UBC.

Second colour: something that has helped when learning got hard.